Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Выполнить приимеры

2 Выполнить дествия по работе с каталогами и файлами

3 Выполнить действия с правами доступа

4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы

Выполнение примеров

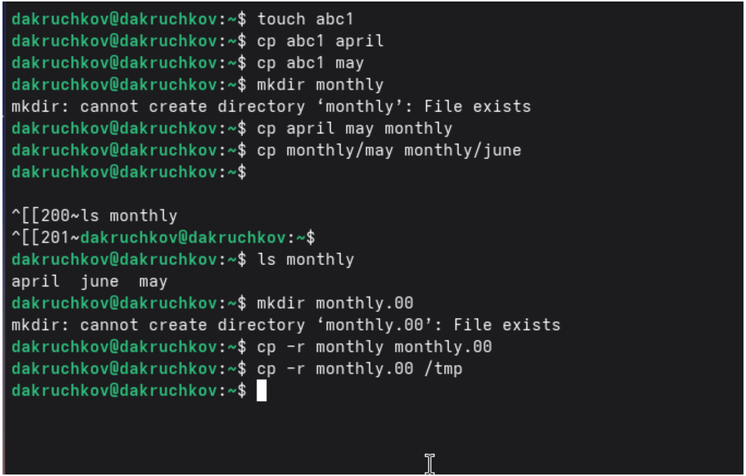


Рисунок 1: Выполнение примеров

Выполнение примеров

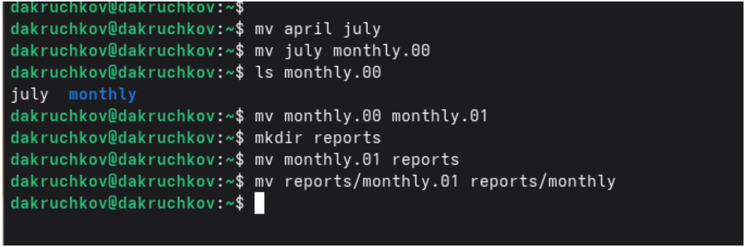
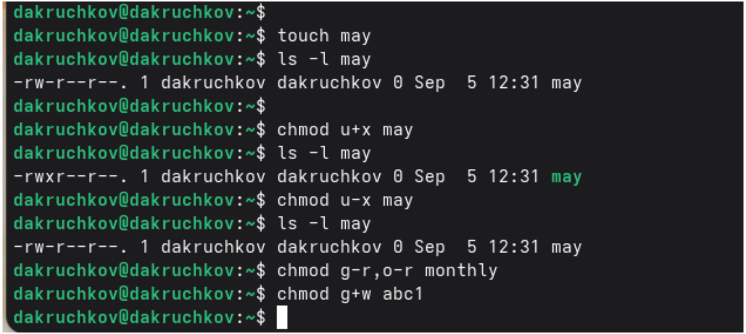


Рисунок 2: Выполнение примеров



```
dakruchkov@dakruchkov:~$
dakruchkov@dakruchkov:~$ touch may
dakruchkov@dakruchkov:~$ ls -l may
-rw-r--r--. 1 dakruchkov dakruchkov 0 Sep  5 12:31 may
dakruchkov@dakruchkov:~$
dakruchkov@dakruchkov:~$ chmod u+x may
dakruchkov@dakruchkov:~$ ls -l may
-rwxr--r--. 1 dakruchkov dakruchkov 0 Sep  5 12:31 may
dakruchkov@dakruchkov:~$ chmod u-x may
dakruchkov@dakruchkov:~$ ls -l may
-rw-r--r--. 1 dakruchkov dakruchkov 0 Sep  5 12:31 may
dakruchkov@dakruchkov:~$ chmod g-r,o-r monthly
dakruchkov@dakruchkov:~$ chmod g+w abc1
dakruchkov@dakruchkov:~$
```

Рисунок 3: Выполнение примеров

Создание директорий и копирование файлов

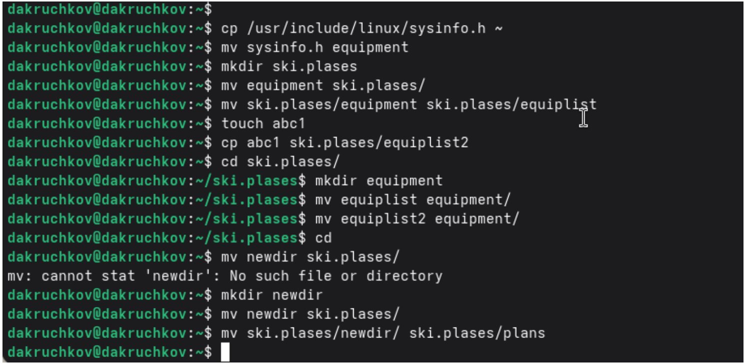
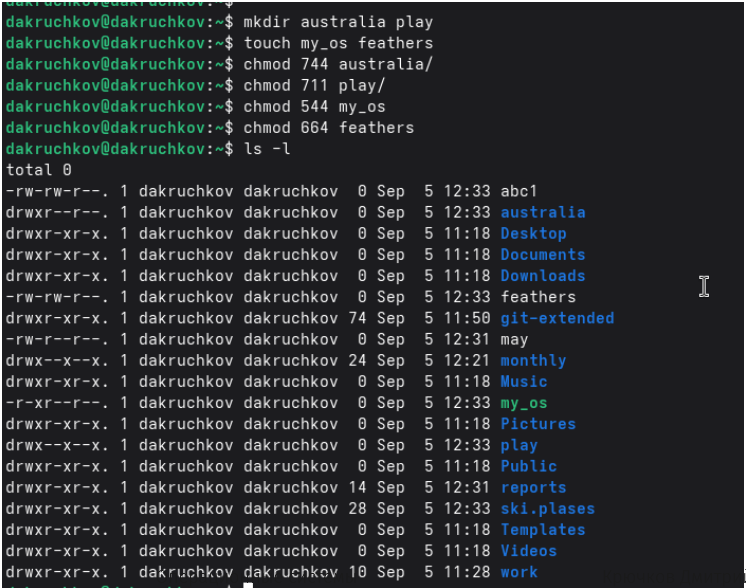
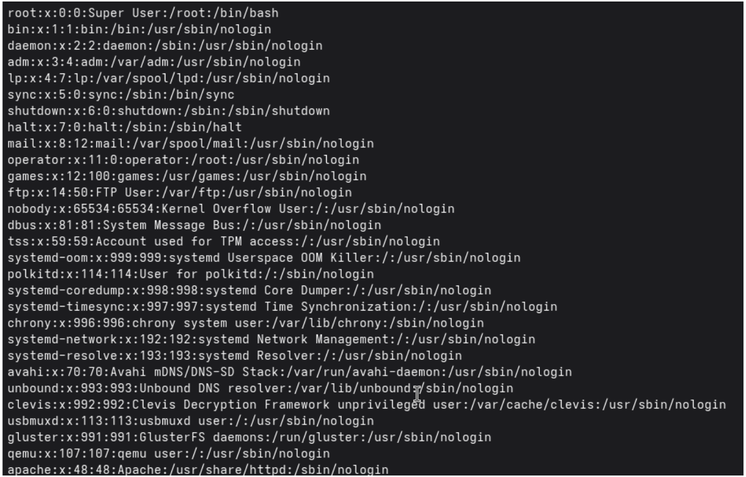


Рисунок 4: Работа с каталогами

Работа с командой chmod



Файл /etc/passwd



```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
```

Рисунок 6: Файл /etc/passwd

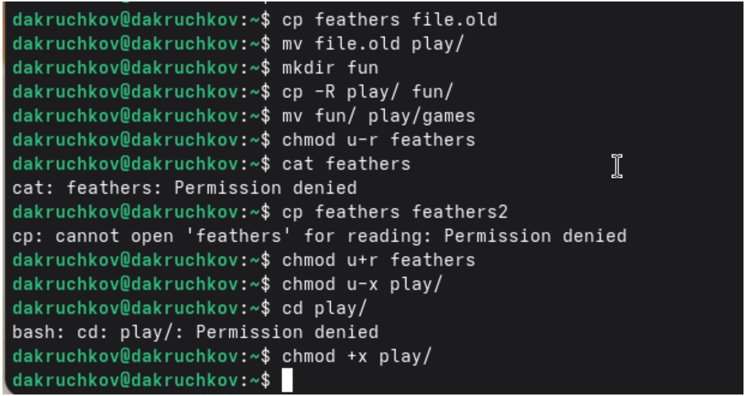


Рисунок 7: Работа с файлами и правами доступа

Справка по командам

```
MOUNT(8)                      System Administration                      MOUNT(8)

NAME
       mount - mount a filesystem

SYNOPSIS
       mount [-h|-V]

       mount [-l] [-t fstype]

       mount -a [-fFnrsvw] [-t fstype] [-O optlist]

       mount [-fnrsvw] [-o options] device|mountpoint

       mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

       mount --bind|--rbind|--move olddir newdir

       mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION
       All files accessible in a Unix system are arranged in one big tree, the file hierarchy,
       rooted at /. These files can be spread out over several devices. The mount command serves to
       attach the filesystem found on some device to the big file tree. Conversely, the umount(8)
       command will detach it again. The filesystem is used to control how data is stored on the
       device or provided in a virtual way by network or other services.

       The standard form of the mount command is:

          mount -t type device dir

       This tells the kernel to attach the filesystem found on device (which is of type type) at the
       directory dir. The option -t type is optional. The mount command is usually able to detect a
       filesystem. The root permissions are necessary to mount a filesystem by default. See section
       "Non-superuser mounts" below for more details. The previous contents (if any) and owner and
       mode of dir become invisible, and as long as this filesystem remains mounted, the pathname
       dir refers to the root of the filesystem on device.
```

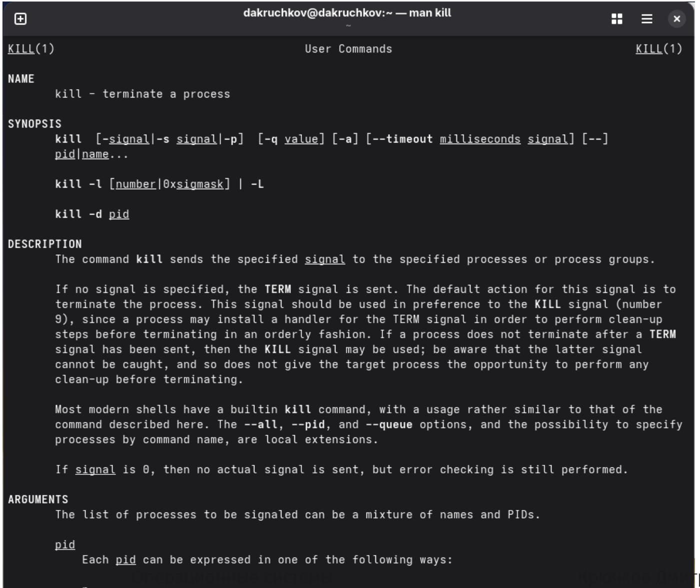
Справка по командам

```
FSCK(8)                        System Administration                        FSCK(8)

NAME
       fsck - check and repair a Linux filesystem

SYNOPSIS
       fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
       fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be
       a device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a
       filesystem label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or
       LABEL=root). Normally, the fsck program will try to handle filesystems on different physical
       disk drives in parallel to reduce the total amount of time needed to check all of them.

       If no filesystems are specified on the command line, and the -A option is not specified, fsck
       will default to checking filesystems in /etc/fstab serially. This is equivalent to the -As
       options.

       The exit status returned by fsck is the sum of the following conditions:

       0
           No errors

       1
           Filesystem errors corrected

       2
           System should be rebooted

       4
           Filesystem errors left uncorrected

       8
           Operational error

       16
```

Справка по командам

```
MKFS(8)                        System Administration                        MKFS(8)

NAME
       mkfs - build a Linux filesystem

SYNOPSIS
       mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
       This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

       mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The
       device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file
       that shall contain the filesystem. The size argument is the number of blocks to be used for
       the filesystem.

       The exit status returned by mkfs is 0 on success and 1 on failure.

       In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype)
       available under Linux. The filesystem-specific builder is searched for via your PATH
       environment setting only. Please see the filesystem-specific builder manual pages for further
       details.

OPTIONS
       -t, --type type
           Specify the type of filesystem to be built. If not specified, the default filesystem type
           (currently ext2) is used.

       fs-options
           Filesystem-specific options to be passed to the real filesystem builder.

       -V, --verbose
           Produce verbose output, including all filesystem-specific commands that are executed.
           Specifying this option more than once inhibits execution of any filesystem-specific
           commands. This is really only useful for testing.

       -h, --help
```

Справка по командам


dakruchkov@dakruchkov:~ — man kill

```
KILL(1)                          User Commands                          KILL(1)

NAME
       kill - terminate a process

SYNOPSIS
       kill  [-signal|-s signal|-p]  [-q value] [-a] [--timeout milliseconds signal] [--]
       pid|name...

       kill -l [number|0xsigmask] | -L

       kill -d pid

DESCRIPTION
       The command kill sends the specified signal to the specified processes or process groups.

       If no signal is specified, the TERM signal is sent. The default action for this signal is to
       terminate the process. This signal should be used in preference to the KILL signal (number
       9), since a process may install a handler for the TERM signal in order to perform clean-up
       steps before terminating in an orderly fashion. If a process does not terminate after a TERM
       signal has been sent, then the KILL signal may be used; be aware that the latter signal
       cannot be caught, and so does not give the target process the opportunity to perform any
       clean-up before terminating.

       Most modern shells have a builtin kill command, with a usage rather similar to that of the
       command described here. The --all, --pid, and --queue options, and the possibility to specify
       processes by command name, are local extensions.

       If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
       The list of processes to be signaled can be a mixture of names and PIDs.

       pid
           Each pid can be expressed in one of the following ways:
```

3. Выводы по проделанной работе

Вывод

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.